

Problem 45 (Canada MO 2015 Problem 1). Let $\mathbb{N}$ be the set of all positive integers. Find all $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$(n-1)^2 < f(n)f(f(n)) < n^2 + n \quad (1)$$

for every positive integer n .

Solution. The only such function is $f(n) = n$ for all $n \in \mathbb{N}$. This is a solution since

$$(n-1)^2 < n^2 < n^2 + n.$$

We prove by induction that $f(n) = n$. Firstly, putting $n = 1$ in (1) gives $0 < f(1)f(f(1)) < 2$. As $f(1)$ and $f(f(1))$ are positive integers, we must have $f(1)f(f(1)) = 1$, and hence $f(1) = f(f(1)) = 1$.

Assume $f(n) = n$ for $n = 1, 2, \dots, k$. Consider any $n \geq k+1$. If $f(n) \leq k$, then $f(f(n)) = f(n)$ by the inductive hypothesis, and so

$$(n-1)^2 < f(n)f(f(n)) = f(n)^2 \leq k^2 \leq (n-1)^2.$$

This is a contradiction. Therefore, we must have $f(n) \geq k+1$ whenever $n \geq k+1$. In particular, we have $f(k+1) \geq k+1$, and hence $f(f(k+1)) \geq k+1$. Note that

$$f(k+1)f(f(k+1)) < (k+1)^2 + (k+1) = (k+1)(k+2).$$

As both $f(k+1)$ and $f(f(k+1))$ are at least $k+1$, the only possibility is $f(k+1) = f(f(k+1)) = k+1$. This proves the inductive step.

By induction, the only solution is $f(n) = n$. □

Problem 46 (Iran MO 2020 2nd Round Problem 5). Find all positive integers n satisfying the following condition: It is possible to partition the set $\{1, 2, \dots, 2n\}$ into n pairs of numbers, such that for each pair (a, b) , the number $ab + 1$ is a perfect square.

Solution. n can be any positive even integer.

If $n = 2m$ for some integer m , we partition the set into the pairs $(1, 3), (2, 4), (5, 7), (6, 8), \dots, (4m-3, 4m-1), (4m-2, 4m)$ so that each pair $(k, k+2)$ satisfies the condition as $k(k+2) + 1 = (k+1)^2$.

If $n = 2m-1$ for some integer m , note that there are m numbers in $S = \{1, 2, \dots, 2n\}$ which are congruent to 2 modulo 4 and $m-1$ numbers which are multiples of 4. Thus, there must be some $a \in S$ with $a \equiv 2 \pmod{4}$ which is paired with a number $b \not\equiv 0 \pmod{4}$. Now, since $ab+1$ is odd, we need $ab+1 \equiv 1 \pmod{8}$ if $ab+1$ is a perfect square. However, this is not true since none of a, b is divisible by 4. So there is no partition satisfying the condition when n is odd. □

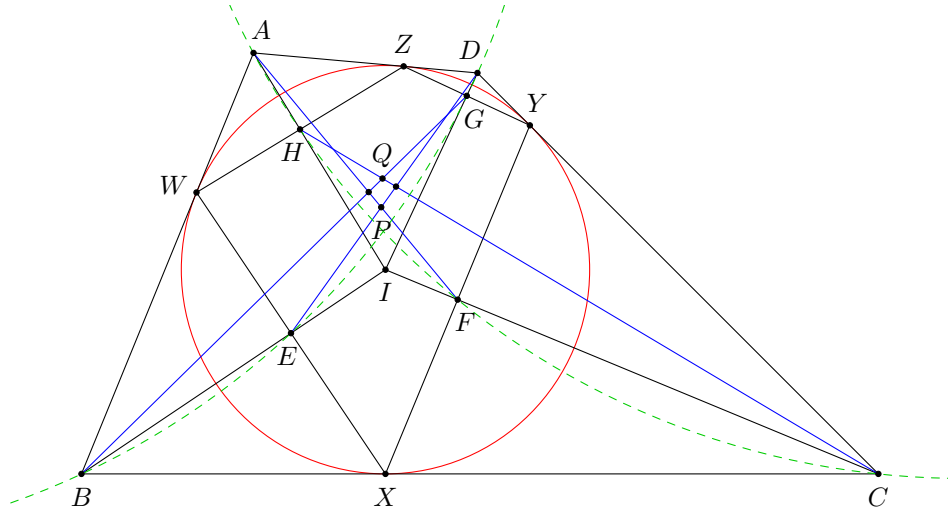
Problem 47 (Circumscribed quadrilateral item 32). Let $ABCD$ be a circumscribed quadrilateral. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively. Let E, F, G, H be the midpoints of WX, XY, YZ, ZW respectively. Prove that AF, BG, CH, DE enclose a cyclic quadrilateral.

Solution. We only provide a proof for the configuration as shown.

Note that AH, BE, CF, DG are concurrent at I since they are the internal angle bisectors of the interior angles of $ABCD$. Also, as $\triangle IHZ \sim \triangle IZA$ and $\triangle IFY \sim \triangle IYC$ by the right angles, we have

$$IH \times IA = IZ^2 = IY^2 = IF \times IC.$$

Therefore, A, H, F, C are concyclic. Similarly, B, E, G, D are concyclic.



Let P be the intersection point of AF and DE . Let Q be the intersection point of BG and CH . Then we have

$$\begin{aligned}
 \angle DPA &= \angle EPF \\
 &= 360^\circ - \angle PEX - \angle PFX - \angle EXF \\
 &= 360^\circ - (90^\circ + \angle IEP) - (90^\circ + \angle IFP) - \angle WXY \\
 &= 180^\circ - \angle IEP - \angle IFP - \angle WXY \\
 &= 180^\circ - (180^\circ - \angle BED) - (180^\circ - \angle AFC) - \angle WXY \\
 &= \angle BED + \angle AFC - \angle WXY - 180^\circ.
 \end{aligned}$$

Similarly, we have $\angle BQC = 540^\circ - \angle BGD - \angle AHC - \angle WZY$. Thus,

$$\angle DPA + \angle BQC = 360^\circ + (\angle BED - \angle BGD) + (\angle AFC - \angle AHC) - (\angle WXY + \angle WZY) = 180^\circ.$$

This shows AF, BG, CH, DE enclose a cyclic quadrilateral. □

Problem 48 (USA TSTST 2015 Problem 6). Let m and n be positive integers, and let S be a finite set of some m -tuples of integers. Consider a game played between two players as follows. Initially, there are m piles of coins put in a row with a fixed order, with the first pile having n coins, and the other piles being empty. The two players take turns choosing an m -tuple $(x_1, x_2, \dots, x_m)$ from S , and add x_j coins to the j th pile for each $j = 1, 2, \dots, m$ (if x_j is negative, adding x_j coins to a pile means removing $-x_j$ coins from that pile, given that the pile has at least $-x_j$ coins). The player who cannot make a valid move will lose the game. Prove that there exist m and S such that the first player has a winning strategy if n is a power of 2, and the second player has a winning strategy if n is not a power of 2.

Solution. We give an example as follows. Consider $m = 9$ and consider the following tuples:

$$\begin{aligned}
 \mathbf{x}_1 &= (-1, 0, 0, 1, 0, 0, 0, 0, 0), \\
 \mathbf{x}_2 &= (-1, 1, 0, -1, 0, 0, 0, 0, 0), \\
 \mathbf{x}_{3j} &= (-j, 0, 0, 0, 0, -1, 0, 0, 1), \quad j = 1, 2, 3, \\
 \mathbf{x}_4 &= (-1, 1, 0, -1, 0, 0, 0, 0, -1), \\
 \mathbf{x}_5 &= (-2, 1, 0, 0, -1, 0, 0, 0, 0), \\
 \mathbf{x}_{6j} &= (-1, -j, 0, 0, -1, 0, 0, 0, 0), \quad j = 0, 1, 2, \\
 \mathbf{x}_{7j} &= (-j, 0, 0, -2, 0, 0, 0, 0, 0), \quad j = 1, 2, \\
 \mathbf{x}_{8j} &= (-j, 0, 0, 0, 0, -2, 0, 0, 0), \quad j = 0, 1, 2, \\
 \mathbf{x}_{9j} &= (-j, 0, 0, -1, -1, 0, 0, 0, 0), \quad j = 0, 1, \\
 \mathbf{x}_{Aj} &= (-j, 0, 0, 0, 0, 0, -1, 0, 0), \quad j = 2, 3, \\
 \mathbf{x}_{Bj} &= (-1, -j, 0, 0, 0, 0, -1, 0, 0), \quad j = 0, 1, 2.
 \end{aligned}$$

For each $\mathbf{x}_i = (a_1, a_2, \dots, a_9)$, define $\mathbf{y}_i = (a_2, a_3, a_1, a_5, a_6, a_4, a_8, a_9, a_7)$ and $\mathbf{z}_i = (a_3, a_1, a_2, a_6, a_4, a_5, a_9, a_7, a_8)$. Let S be the set containing all these $\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i$. We claim that this set satisfies the conditions. In the following, we use $\mathbf{v}$ to denote the current tuple of coins.

Claim 1: If $\mathbf{v} = (a, b, 0, 0, \dots, 0)$ where a is odd, then the first player has a winning strategy if $(a, b) = (1, 0)$, and the second player has a winning strategy if not.

Proof. We prove this by induction on $a + b$. For the base case $a + b = 1$ (which means $(a, b) = (1, 0)$), the first player can choose $\mathbf{x}_1$ so that $\mathbf{v} = (0, 0, 0, 1, 0, 0, 0, 0, 0)$. The next player loses since all tuples in S decreases some of the first three entries, or decreases the middle three entries by 2 (e.g. $\mathbf{x}_{80}, \mathbf{x}_{90}$).

For the inductive step, the first player only has two possible moves since only $\mathbf{x}_1$ and its variants does not decrease any of the last six entries.

- If the first player chooses $\mathbf{x}_1$, then $\mathbf{v} = (a - 1, b, 0, 1, 0, 0, 0, 0, 0)$.
 - If $a \geq 3$, the second player can choose $\mathbf{x}_2$ so that $\mathbf{v} = (a - 2, b + 1, 0, 0, \dots, 0)$. Since $a - 2$ is odd, $b + 1 \neq 0$ and $(a - 2) + (b + 1) < a + b$, the second player has a winning strategy by the inductive hypothesis.
 - If $a = 1$ (hence $b \neq 0$), then $\mathbf{v} = (0, b, 0, 1, 0, 0, 0, 0, 0)$. This is essentially the same as $\mathbf{v} = (b, 0, 0, 0, 0, 1, 0, 0, 0)$ by the cyclic symmetry. The second player can choose $\mathbf{x}_{3j}$ such that $c := b - j \in \{0, 3, 5, 7, \dots\}$ (since there are 3 consecutive choices of j and $b \neq 0$). Then we have $\mathbf{v} = (c, 0, 0, 0, 0, 0, 0, 0, 1)$. The only possible move for the first player is to choose $\mathbf{x}_1$ (when $c > 0$). Then we have $\mathbf{v} = (c - 1, 0, 0, 1, 0, 0, 0, 0, 1)$. Now, the second player can choose $\mathbf{x}_4$ so that $\mathbf{v} = (c - 2, 1, 0, 0, \dots, 0)$. Since $c - 2$ is odd and $(c - 2) + 1 = c - 1 < a + b$, we are done by the inductive hypothesis.
- If the first player chooses $\mathbf{z}_1$, then $\mathbf{v} = (a, b - 1, 0, 0, 1, 0, 0, 0, 0)$ (note that $b \neq 0$).
 - If $a \geq 3$, the second player can choose $\mathbf{x}_5$ so that $\mathbf{v} = (a - 2, b, 0, 0, \dots, 0)$. Since $a - 2$ is odd, $b \neq 0$ and $(a - 2) + b < a + b$, the second player has a winning strategy by the inductive hypothesis.
 - If $a = 1$ so that $\mathbf{v} = (1, b - 1, 0, 0, 1, 0, 0, 0, 0)$, the second player can choose $\mathbf{x}_{6j}$ such that $c := b - 1 - j \in \{0, 3, 5, 7, \dots\}$. Then we have $\mathbf{v} = (0, c, 0, 0, \dots, 0)$. This is the same as $\mathbf{v} = (c, 0, 0, \dots, 0)$. As either $c = 0$ or $c \geq 3$ is odd (and $c < a + b$), we are done by the inductive hypothesis.

This proves Claim 1. □

Claim 2: If $\mathbf{v} = (a, b, 0, 0, \dots, 0)$ where a is even, $a \geq 2$, and $a + 2b$ is not a power of 2, then the second player has a winning strategy.

Proof. We prove this by induction on $a + b$ and a (which means the case (a, b) preceeds (a', b') if $a + b < a' + b'$, or $a + b = a' + b'$ and $a < a'$). The base case can be combined with the inductive step. The first player only has two possible moves.

- If the first player chooses $\mathbf{x}_1$, then $\mathbf{v} = (a - 1, b, 0, 1, 0, 0, 0, 0, 0)$. The second player can choose $\mathbf{x}_2$ so that $\mathbf{v} = (a - 2, b + 1, 0, 0, \dots, 0)$.
 - If $a \geq 4$, then $a - 2$ is even, $a - 2 \geq 2$ and $(a - 2) + 2(b + 1) = a + 2b$. So we are done by the inductive hypothesis.
 - If $a = 2$ and b is even (hence $b \neq 0$), then $\mathbf{v} = (0, b + 1, 0, 0, \dots, 0)$. This is the same as $(b + 1, 0, 0, \dots, 0)$ where $b + 1$ is odd and $b + 1 \neq 1$. We are done by Claim 1.
 - If $a = 2$ and b is odd, then $\mathbf{v} = (0, b + 1, 0, 0, \dots, 0)$. This is the same as $(b + 1, 0, 0, \dots, 0)$ where $b + 1$ is even, $b + 1 \geq 2$, and $(b + 1) + 2(0) = b + 1$ is not a power of 2 (since $a + 2b = 2(1 + b)$ is not a power of 2). Also, we have $(b + 1) + 0 < a + b$. So we are done by the inductive hypothesis.
- If the first player chooses $\mathbf{z}_1$, then $\mathbf{v} = (a, b - 1, 0, 0, 1, 0, 0, 0, 0)$. The second player can choose $\mathbf{x}_{60}$ so that $\mathbf{v} = (a - 1, b - 1, 0, 0, \dots, 0)$. Since $a - 1$ is odd and $(a - 1, b - 1) \neq (1, 0)$ (otherwise $a + 2b = 4$), we are done by Claim 1.

This proves Claim 2. □

Claim 3: If $\mathbf{v} = (a, b, 0, 0, \dots, 0)$ where a is even, $a \geq 2$, and $a + 2b$ is a power of 2, then the first player has a winning strategy.

Proof. We prove this by induction on $a + b$ and a . The base case can be combined with the inductive step. The first player can choose $\mathbf{x}_1$ so that $\mathbf{v} = (a - 1, b, 0, 1, 0, 0, 0, 0, 0)$. The second player has four possible moves.

- If the second player chooses $\mathbf{x}_1$, then $\mathbf{v} = (a - 2, b, 0, 2, 0, 0, 0, 0, 0)$.

- If $a \geq 4$ and $(a, b) \neq (4, 0)$, the first player can choose $\mathbf{x}_{71}$ so that $\mathbf{v} = (a - 3, b, 0, 0, \dots, 0)$. Since $a - 3$ is odd and $(a - 3, b) \neq (1, 0)$, we are done by Claim 1.
 - If $a = 4$ and $b = 0$, then $\mathbf{v} = (2, 0, 0, 2, 0, 0, 0, 0, 0)$. The first player can choose $\mathbf{x}_{72}$ so that $\mathbf{v} = \mathbf{0}$. The second player loses.
 - If $a = 2$, then $\mathbf{v} = (0, b, 0, 2, 0, 0, 0, 0, 0)$. This is the same as $\mathbf{v} = (b, 0, 0, 0, 0, 2, 0, 0, 0)$. The first player can choose $\mathbf{x}_{8j}$ such that $c := b - j \in \{0, 3, 5, 7, \dots\}$. Then we have $\mathbf{v} = (c, 0, 0, \dots, 0)$. As either $c = 0$ or $c \geq 3$ is odd, we are done by Claim 1.
- If the second player chooses $\mathbf{z}_1$, then $\mathbf{v} = (a - 1, b - 1, 0, 1, 1, 0, 0, 0, 0)$.
 - If $(a, b) \neq (2, 1)$, the first player can choose $\mathbf{x}_{90}$ so that $\mathbf{v} = (a - 1, b - 1, 0, 0, \dots, 0)$. Since $a - 1$ is odd and $(a - 1, b - 1) \neq (1, 0)$, we are done by Claim 1.
 - If $(a, b) = (2, 1)$, then $\mathbf{v} = (1, 0, 0, 1, 1, 0, 0, 0, 0)$. The first player can choose $\mathbf{x}_{91}$ so that $\mathbf{v} = \mathbf{0}$. The second player loses.
 - If the second player chooses $\mathbf{x}_2$, then $\mathbf{v} = (a - 2, b + 1, 0, 0, \dots, 0)$. Since $a - 2$ is even and $(a - 2) + 2(b + 1) = a + 2b$, we are done by the inductive hypothesis except $a = 2$. When $a = 2$, it is the same as $\mathbf{v} = (b + 1, 0, 0, \dots, 0)$. Note that $b + 1$ is a power of 2 since $a + 2b = 2(1 + b)$ is a power of 2. Thus, we are done by the inductive hypothesis or by Claim 1 (when $b = 0$).
 - If the second player chooses $\mathbf{z}_{3j}$ where $j = 1, 2, 3$, then $\mathbf{v} = (a - 1, b - j, 0, 0, 0, 0, 1, 0, 0)$.
 - If $a \geq 4$ and $(a, b - j) \neq (4, 0)$, the first player can choose $\mathbf{x}_{A2}$ so that $\mathbf{v} = (a - 3, b - j, 0, 0, \dots, 0)$. Since $a - 3$ is odd and $(a - 3, b - j) \neq (1, 0)$, we are done by Claim 1.
 - If $a = 4$ and $b - j = 0$, then $\mathbf{v} = (3, 0, 0, 0, 0, 0, 1, 0, 0)$. The first player can choose $\mathbf{x}_{A3}$ so that $\mathbf{v} = \mathbf{0}$. The second player loses.
 - If $a = 2$, then $\mathbf{v} = (1, b - j, 0, 0, 0, 0, 1, 0, 0)$. The first player can choose $\mathbf{x}_{Bk}$ such that $c := b - j - k \in \{0, 3, 5, 7, \dots\}$. Then we have $\mathbf{v} = (0, c, 0, 0, \dots, 0)$, which is the same as $\mathbf{v} = (c, 0, 0, \dots, 0)$. As either $c = 0$ or $c \geq 3$ is odd, we are done by Claim 1.

This proves Claim 3. □

The whole proof is complete by Claim 2 and Claim 3. □